Cutoff Sensitivity of Strict Splitting Counts
An Exact Extension to 5000 on the Fixed Curve Cohort

Anonymous 30 October 2026

Abstract

We extend the previously reported strict Frobenius-discriminant splitting-count comparison to every integer cutoff from 100 through 5000, retaining all 21 curve rows and enumerating all 4480 original stream-count-preserving label assignments. The observed within-field separation occurs only at $691 \le L \le 718$ and $769 \le L \le 772$: 32 of 4901 integer cutoffs (0.6529%). No cutoff above 772 separates the paired publications and named unpublished candidates. A separate native implementation reconstructs all five output files; three deliberate corruptions are refused. This is retrospective exact reanalysis, with no new curves or point counts. It establishes cutoff sensitivity in this cohort, not a historical selection rule, unusualness against random curves, or an ECDLP advantage.

1 Question and fixed method

The original degree-691 observation was reported before this work [1, 2]. For each inherited cofactor-one curve, with certified full order n, set

$$t = p + 1 - n, \qquad \Delta = t^2 - 4p, \qquad S_L = \#\{\text{odd primes } \ell \le L : (\Delta/\ell) = +1\}.$$

We retain ramified characters separately as R_L and report $S_L + R_L$ only as a sensitivity analysis. Prime 2 is exceptional and has no computed Legendre character. The count is a property of p, n; changing the base point does not change it. Background on Elkies/Atkin characters does not provide a finite conditional random-curve model for these inputs [3].

For each of the four fields with both labels, define $m_L = \min_{\text{unpublished}} S_L - \max_{\text{published}} S_L$. The joint rule succeeds exactly when all four margins are positive. The 384-/521-bit publications remain in the data without invented same-field comparison margins. Every integer cutoff is retained; 645 effective positions represent the requested range.

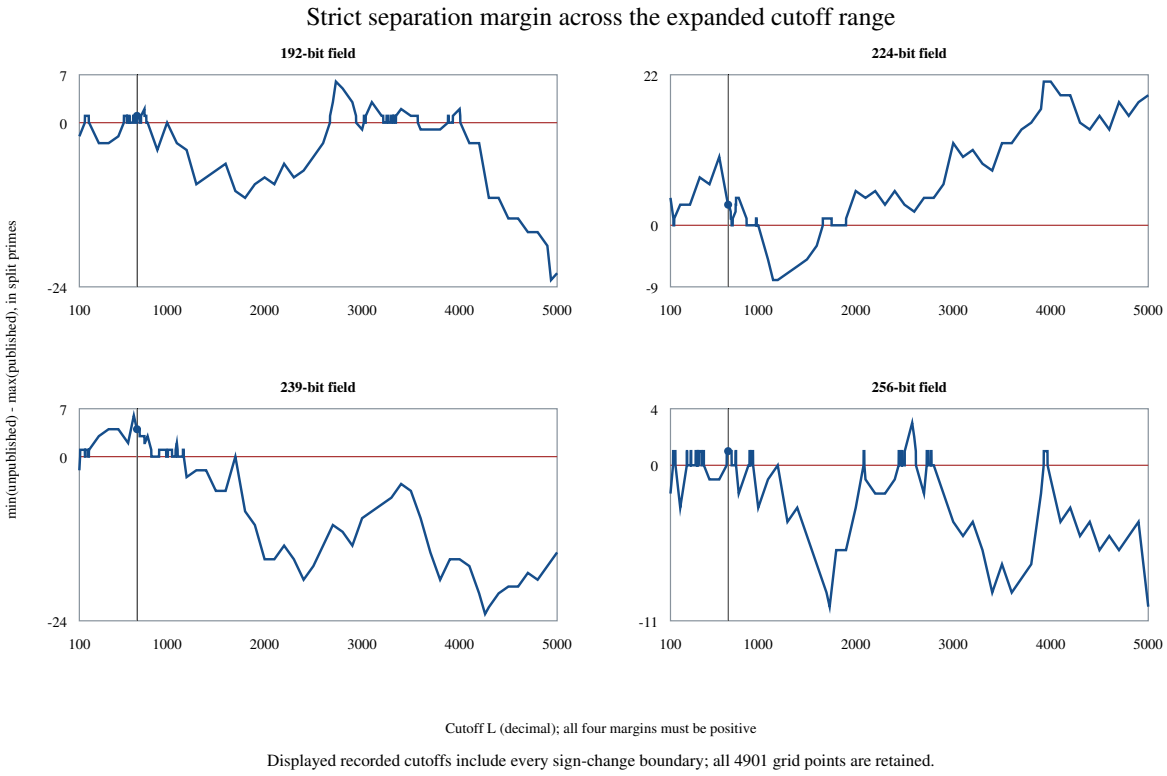


Figure 1: Recorded strict margins in the fixed cohort. The horizontal line is zero; the vertical marker is 691. Display points include a 100-step grid, 691, all sign-change boundaries, extrema and joint-success boundaries. Lines guide the eye; all 4901 integer positions are retained in the raw data. These are exact finite counts, so sampling-error bars are inapplicable.

2 Full-range results

The only strict success intervals are $[691, 718]$ and $[769, 772]$, inclusive. They contain 32 integer cutoffs and four effective prime cutoffs: 691, 701, 709 and 769. Thus the ordering holds at 0.6529% of the integer grid and at 0.6202% of effective positions. It fails at every cutoff from 773 through 5000. Including ramified primes yields *zero* joint-success cutoffs in the requested range.

Field bits	Positive cutoffs	Percent	m_{691}	m_{5000}
192	1018	20.7713	1	-22
224	3957	80.7386	3	19
239	862	17.5882	4	-14
256	374	7.6311	1	-10

Table 1: Within-field strict ordering across all 4901 integer cutoffs. A positive margin means every paired publication has a smaller count than every named unpublished candidate in that field. The percentages describe this grid, not a sampling probability or statistical comparison between fields.

P-224 is the more persistent case within this cohort. At 5000 its publication has strict count 299, compared with 318, 342 and 325 for its three controls. The other three paired fields have negative margins at 5000. Their publication/control ordering therefore reverses or overlaps. This does not classify any curve as cryptographically weak.

2.1 Label enumeration and the cutoff-family boundary

All 4480 assignments preserve the number of publications in each original stream. The requested 100–5000 family allows a strict separator for 29/4480 assignments (0.6473%). The separate inclusive family allows one for 30/4480 assignments, even though the *observed* inclusive labels never separate.

Strict cutoff family	Effective positions	Successful assignments
Original, 2–2000	303	23/4480
Requested, 100–5000	645	29/4480
Supplementary union, 2–5000	669	30/4480

The first two families are not nested: the second removes cutoffs below 100. Exactly 22 successful assignments are shared, seven occur only in the requested family, and one only in the old family. The supplementary 30 count is their set union, computed descriptively from the two recorded flags; it is not a new prospective test. The fixed-691 count remains 1/4480.

All these proportions are retrospective restricted-family enumerations. They do not establish historical exchangeability, correct for every earlier trait search, or give a probability that a designer used a backdoor.

3 Interpretation

The result supports a narrow, cutoff-sensitive pattern in the named cohort. It neither authenticates historical candidate visits or rejections nor supplies an attack. Known isogeny-based DLP reductions require their own hypotheses and a destination-solver advantage [4]; this count supplies neither. The broader 69-position draft-compatible census has not been classified by this reanalysis. The requested 10,000 accepted random curves per field remain unrun. This loop adds no test of a hidden base-point–b relationship.

4 Every retained curve at 691 and 5000

		$L = 691$			$L = 5000$		
Curve	Role	S	R	$S + R$	S	R	$S + R$
p192-139	P	60	3	63	335	3	338
p192-2198	U	61	0	61	339	1	340
p192-5228	P	57	1	58	317	1	318
p192-5405	U	62	0	62	313	1	314
p192-6013	P	58	1	59	303	2	305
p224-777	U	53	4	57	318	4	322
p224-1485	U	56	3	59	342	3	345
p224-5736	U	59	2	61	325	3	328
p224-7026	P	50	3	53	299	3	302
p239-36	U	71	2	73	347	2	349
p239-1562	U	67	3	70	337	3	340
p239-2185	P	42	1	43	315	1	316
p239-2320	P	57	2	59	320	2	322
p239-2491	U	63	2	65	352	2	354
p239-2682	U	61	4	65	322	4	326
p239-2863	U	66	1	67	329	1	330
p239-3048	P	57	1	58	336	1	337
p256-876	U	58	2	60	318	4	322
p256-1842	P	57	2	59	328	2	330
p384-2688	P	54	0	54	324	0	324
p521-1136	P	52	1	53	339	1	340

Table 2: All 21 rows. P means published and U means a named unpublished candidate; U does not certify historical visitation or rejection. S counts strict positive characters and R counts ramified characters. No singleton control or unknown historical base point is used as a numeric rejection label.

The NIST P-192 row is `p192-139`; the other two published 192-bit rows are `prime192v3` (5228) and `prime192v2` (6013). The published 239-bit rows 2185, 2320 and 3048 are the three X9.62 examples. All orders, seeds and coefficient branches are inherited unchanged.

Provenance and execution status

Source, fixed input and commands were committed before execution at `93501d9143bf56cfa3f5cd1abb528b1f48daef14`. The input SHA-256 begins `f425dc7c323bdbe6`; full bindings are in `BUILD.json` and `SOURCES-SHA256SUMS`. Run identifier: `nist-expanded-cutoff-reanalysis-run-001`.

This was local deterministic reanalysis of existing certified results under the repository's existing-results development exception. It was not a new sampling experiment, point count or performance benchmark. Each native child had a 1536 MiB address-space cap and a fixed timeout; no timing is admitted. All ten expected phase outcomes matched. No S3 checkpoint or remote raw-artifact receipt is claimed. The full raw records, failed presentation attempts and development logs are preserved in a separate downloadable local archive. The earlier paper and its frozen numerical records remain unchanged.

5 Verification and record boundaries

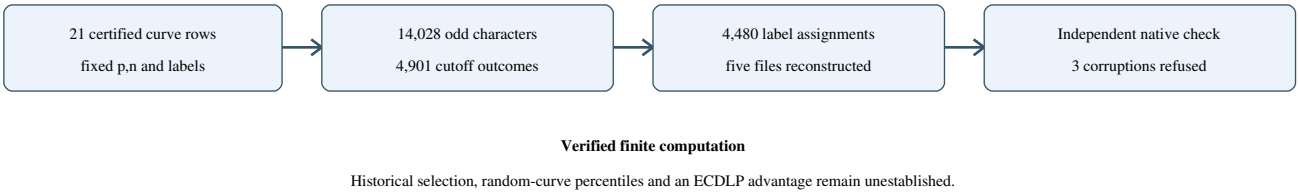


Figure 2: Scope of the independently checked finite calculation.

The native GMP producer and separate PARI verifier agree byte-for-byte on five files, covering 14,028 odd-prime characters, 102,921 curve/cutoff rows, all 4901 cutoff outcomes and all 4480 label assignments. The three altered character/calibration/summary outputs are each refused with native exit 1. The implementations may share GMP primitives.

References

[1] asanso. Prior strict degree-691 observation, 17 October 2026. `https://x.com/asanso/status/2111511837866893580`. Provenance is the source-bound retrieval recorded in the earlier replication.

[2] Anonymous. *Strict Splitting Counts in Reconstructed Elliptic-Curve Candidate Streams: An Exact Reproduction and Its Limits*. Repository draft, 29 October 2026; not submitted. Source paper commit `b3e7c31dd70c123d5ba36e5a335c660c40c52d3f`.

[3] I. E. Shparlinski and A. V. Sutherland. *On the Distribution of Atkin and Elkies Primes*. arXiv:1112.3390v2, 2011.

[4] D. Jao, S. D. Miller and R. Venkatesan. *Expander graphs based on GRH with an application to elliptic curve cryptography*. arXiv:0811.0647v2, 2009.